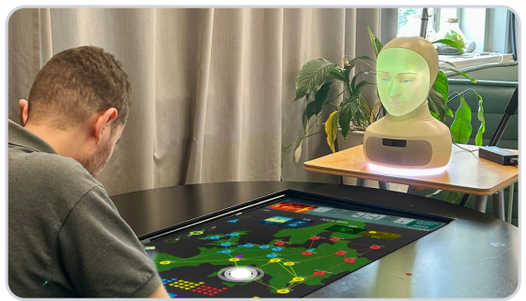
1 Testbed



state ↓ ↑ moves

rule-based solver

HRI corpus
70 sessions · 36 participants
speech · video · gaze · game logs

2 Partner signal

training time only · predicted (RQ1)

Self-reported

enjoyment ●●●●○
rated once, after each game

Speech-to-Joy

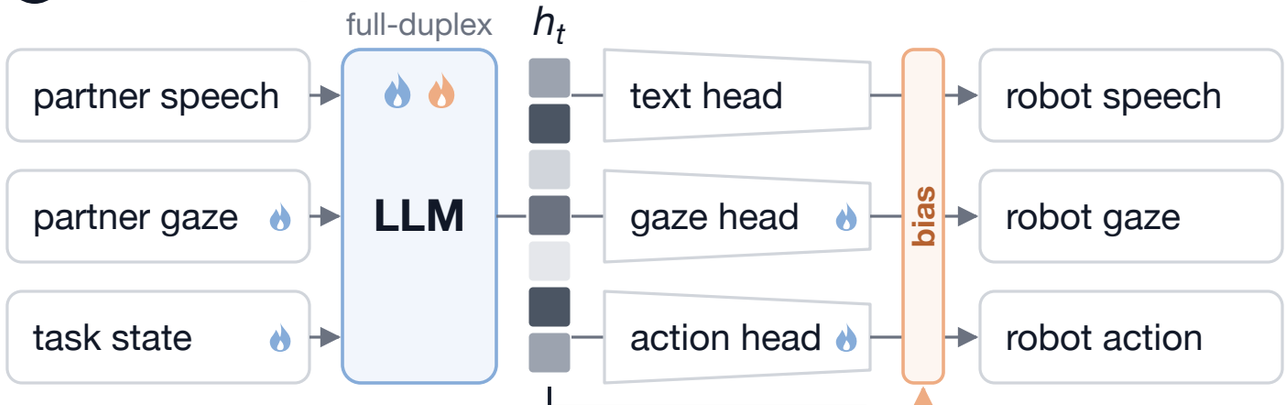


speech → *predicted enjoyment*

in real time, from speech alone
outperforms an HRI annotator

3 Model adaptation *current/future work*

trained



stage 1 · ground
$\pi \cdot v$ self-play
conversational gaze
general task robot · *all heads fine-tuned on abundant data*

stage 2a · imitate
intervention head
task-specific robot ❶

stage 2b · RL
reward from predicted state ❷
partner-specific robot